

Foundations of machine learning

## Bonus Sides: Reproducing Kernel Hilbert Spaces and Splines

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# Splines and Reproducing Kernel Hilbert Spaces

- Penalized least squares: For some (semi-)norm  $\|f\|$ ,

$$\hat{f} = \operatorname{argmin}_f \sum_i (Y_i - f(X_i))^2 + \lambda \|f\|^2.$$

- Leading case: Splines, e.g.,

$$\hat{f} = \operatorname{argmin}_f \sum_i (Y_i - f(X_i))^2 + \lambda \int f''(x)^2 dx.$$

- Can we think of penalized regressions in terms of a prior?
- If so, what is the prior distribution?

## The finite dimensional case

- Consider the finite dimensional analog to penalized regression:

$$\hat{\theta} = \operatorname{argmin}_t \sum_{i=1}^n (X_i - t_i)^2 + \|t\|_C^2,$$

where

$$\|t\|_C^2 = t' C^{-1} t.$$

- We saw before that this is the posterior mean when
  - $X|\theta \sim N(\theta, I_k),$
  - $\theta \sim N(0, C).$

## The reproducing property

- The norm  $\|t\|_C$  corresponds to the inner product

$$\langle t, s \rangle_C = t' C^{-1} s.$$

- Let  $C_i = (C_{i1}, \dots, C_{ik})'$ .
- Then, for any vector  $y$ ,

$$\langle C_i, y \rangle_C = y_i.$$

### Practice problem

Verify this.

## Reproducing kernel Hilbert spaces

- Now consider a general Hilbert space of functions equipped with an inner product  $\langle \cdot, \cdot \rangle$  and corresponding norm  $\| \cdot \|$ ,
- such that for all  $x$  there exists an  $M_x$  such that for all  $f$

$$f(x) \leq M_x \cdot \|f\|.$$

- Read: “Function evaluation is continuous with respect to the norm  $\| \cdot \|$ .”
- Hilbert spaces with this property are called reproducing kernel Hilbert spaces (RKHS).
- Note that  $L^2$  spaces are not RKHS in general!

# The reproducing kernel

- Riesz representation theorem:  
For every continuous linear functional  $L$  on a Hilbert space  $\mathcal{H}$ ,  
there exists a  $g_L \in \mathcal{H}$  such that for all  $f \in \mathcal{H}$

$$L(f) = \langle g_L, f \rangle.$$

- Applied to function evaluation on RKHS:

$$f(x) = \langle C_x, f \rangle$$

- Define the reproducing kernel:

$$C(x_1, x_2) = \langle C_{x_1}, C_{x_2} \rangle.$$

- By construction:

$$C(x_1, x_2) = C_{x_1}(x_2) = C_{x_2}(x_1)$$

## Practice problem

- Show that  $C(\cdot, \cdot)$  is positive semi-definite, i.e., for any  $(x_1, \dots, x_k)$  and  $(a_1, \dots, a_k)$

$$\sum_{i,j} a_i a_j C(x_i, x_j) \geq 0.$$

- Given a positive definite kernel  $C(\cdot, \cdot)$ , construct a corresponding Hilbert space.

## Solution

- Positive definiteness:

$$\begin{aligned}\sum_{i,j} a_i a_j C(x_i, x_j) &= \sum_{i,j} a_i a_j \langle C_{x_i}, C_{x_j} \rangle \\ &= \left\langle \sum_i a_i C_{x_i}, \sum_j a_j C_{x_j} \right\rangle = \left\| \sum_i a_i C_{x_i} \right\|^2 \geq 0.\end{aligned}$$

- Construction of Hilbert space: Take linear combinations of the functions  $C(x, \cdot)$  (and their limits) with inner product

$$\left\langle \sum_i a_i C(x_i, \cdot), \sum_j b_j C(y_j, \cdot) \right\rangle_C = \sum_{i,j} a_i b_j C(x_i, y_j).$$



- Kolmogorov consistency theorem:  
For a positive definite kernel  $C(\cdot, \cdot)$   
we can always define a corresponding prior

$$f \sim GP(0, C).$$

- Recap:
  - For each regression penalty,
  - when function evaluation is continuous w.r.t. the penalty norm
  - there exists a corresponding prior.
- Next:
  - The solution to the penalized regression problem
  - is the posterior mean for this prior.

## Solution to penalized regression

- Let  $f$  be the solution to the penalized regression

$$\hat{f} = \operatorname{argmin}_f \sum_i (Y_i - f(X_i))^2 + \lambda \|f\|_C^2.$$

### Practice problem

- Show that the solution to the penalized regression has the form

$$\hat{f}(x) = c(x) \cdot (C + n\lambda I)^{-1} \cdot Y,$$

where  $C_{ij} = C(X_i, X_j)$  and  $c(x) = (C(X_1, x), \dots, C(X_n, x))$ .

- Hints

- Write  $\hat{f}(\cdot) = \sum a_i \cdot C(X_i, \cdot) + \rho(\cdot)$ ,
- where  $\rho$  is orthogonal to  $C(X_i, \cdot)$  for all  $i$ .
- Show that  $\rho = 0$ .
- Solve the resulting least squares problem in  $a_1, \dots, a_n$ .

## Solution

- Using the reproducing property, the objective can be written as

$$\begin{aligned} & \sum_i (Y_i - f(X_i))^2 + \lambda \|f\|_C^2 \\ &= \sum_i (Y_i - \langle C(X_i, \cdot), f \rangle)^2 + \lambda \|f\|_C^2 \\ &= \sum_i \left( Y_i - \left\langle C(X_i, \cdot), \sum_j a_j \cdot C(X_j, \cdot) + \rho \right\rangle \right)^2 + \lambda \left\| \sum_i a_i \cdot C(X_i, \cdot) + \rho \right\|_C^2 \\ &= \sum_i \left( Y_i - \sum_j a_j \cdot C(X_i, X_j) \right)^2 + \lambda \left( \sum_{i,j} a_i a_j C(x_i, x_j) + \|\rho\|_C^2 \right) \\ &= \|Y - C \cdot a\|^2 + \lambda (a' C a + \|\rho\|_C^2) \end{aligned}$$

- Given  $a$ , this is minimized by setting  $\rho = 0$ .
- Now solve the quadratic program using first order conditions.

# Splines

- Now what about the spline penalty

$$\int f''(x)^2 dx?$$

- Is function evaluation continuous for this norm?
- Yes, if we restrict to functions such that  $f(0) = f'(0) = 0$ .
- The penalty is a semi-norm that equals 0 for all linear functions.
- It corresponds to the GP prior with

$$C(x_1, x_2) = \frac{x_1 x_2^2}{2} - \frac{x_2^3}{6}$$

for  $x_2 \leq x_1$ .

- This is in fact the covariance of integrated Brownian motion!

## Practice problem

Verify that  $C$  is indeed the reproducing kernel for the inner product

$$\langle f, g \rangle = \int_0^1 f''(x)g''(x)dx.$$

- Takeaway: Spline regression is equivalent to the limit of a posterior mean where the prior is such that

$$f(x) = A_0 + A_1 \cdot x + g$$

where

$$g \sim GP(0, C)$$

and

$$A \sim N(0, \nu \cdot I)$$

as  $\nu \rightarrow \infty$ .

## Solution

- Have to show:  $\langle C_x, g \rangle = g(x)$
- Plug in definition of  $C_x$
- Last 2 steps: use integration by parts, use  $g(0) = g'(0) = 0$
- This yields:

$$\begin{aligned}\langle C_x, g \rangle &= \int C_x''(y) g''(y) dy \\&= \int_0^x \left( \frac{xy^2}{2} - \frac{y^3}{6} \right)'' g''(y) dy + \int_x^1 \left( \frac{yx^2}{2} - \frac{x^3}{6} \right)'' g''(y) dy \\&= \int_0^x (x-y) g''(y) dy \\&= x \cdot (g'(x) - g'(0)) + \int_0^x g'(y) dy - (yg'(y)) \Big|_{y=0}^x \\&= g(x).\end{aligned}$$

# References

- Gaussian process priors:

*Williams, C. and Rasmussen, C. (2006). Gaussian processes for machine learning. MIT Press, chapter 2.*

- Splines and Reproducing Kernel Hilbert Spaces

*Wahba, G. (1990). Spline models for observational data, volume 59. Society for Industrial Mathematics, chapter 1.*